// STRATEGY RESEARCH NOTE

R2VAULT VOLATILITY BREAKOUT

A long-only NQ futures strategy on 5-minute bars. It defines a vault during weak linear structure and trades a filtered breakout in the direction of positive slope.

INSTRUMENT	TIMEFRAME	SAMPLE	PERIOD
@NQ# futures	**5 minutes**	**2,518 trades**	**2017 / 2026**

// CORE HYPOTHESIS

FROM NOISE TO BREAKOUT.

R2Vault uses a rolling linear regression to separate two different questions: how well a line describes recent prices, and which direction that line points. Low R2 creates the reference structure; positive slope selects the long side.

The development deliberately remained limited to two strategy versions. No dedicated parameter optimization was performed on the out-of-sample segment, reducing opportunities to fit historical noise.

// IMPORTANT SCOPE

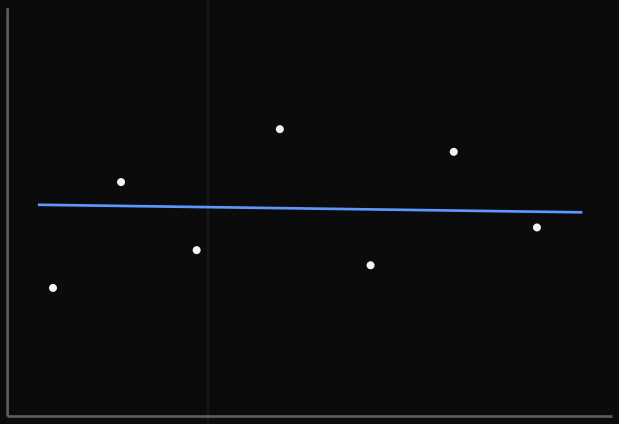
This is a systematic research model based on closing prices. It does not use order flow, volume profile, or a forward-looking volatility forecast.

// WORKFLOW

R2 near zero stores the candle range. Positive slope enables a buy stop above that range plus a proportional noise buffer.

// REGRESSION MECHANICS

R2 AND SLOPE.



// LOW R2: WEAK EXPLANATORY FIT

R2 is the coefficient of determination over the last 576 closes. It measures the fraction of price variance explained by the regression line: 0 means no linear explanatory power, while 1 is a perfect linear fit. R2 has no direction because it is squared.

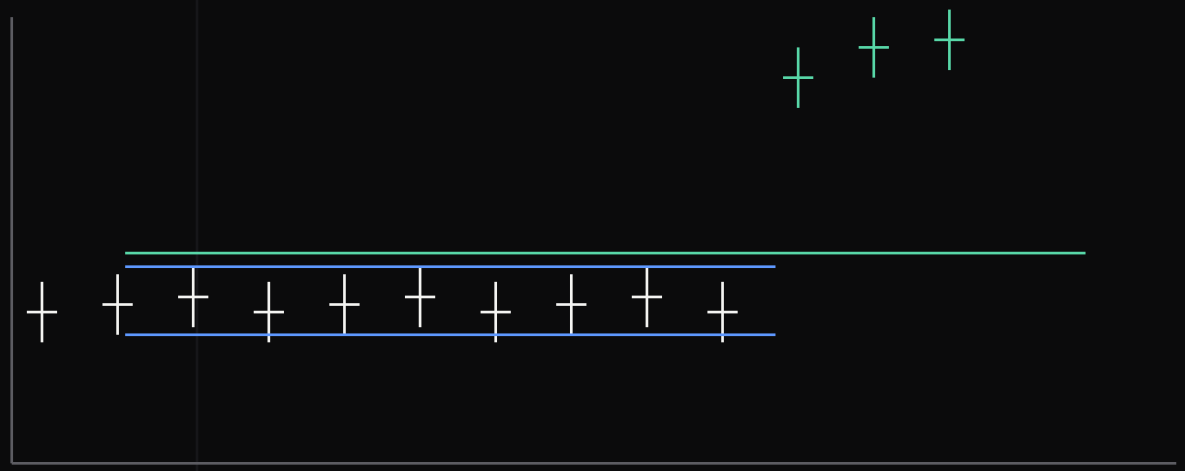
Slope is the direction of the fitted line. In this implementation the most recent bars receive higher X values, so Slope > 0 means the fitted relationship rises toward the present. R2 answers fit quality; Slope answers direction.

// OFFICIAL THRESHOLD

R2ZeroThreshold = 0.030. A vault is created when the line explains no more than 3% of observed variation: an extremely weak linear structure.

// STRUCTURE AND ENTRY

THE VAULT.



// ENTRYLEVEL = VAULTHIGH + (VAULTRANGE X 0.20)

// VAULT

If R2 <= 0.030, the current High and Low replace the stored vault. VaultRange is High minus Low.

// FILTERED BREAKOUT

The buy stop is placed above VaultHigh plus 20% of VaultRange, only in-session, with Slope > 0, and at most three entries per day.

// OFFICIAL LIVE INPUTS

PARAMETERS USED.

PRICE **Close**	R2LENGTH **576**
R2ZEROTHRESHOLD **0.030**	R2TAKEPROFITTHRESHOLD **2.00**
SESSION **09:30 - 15:30**	MAXTRADESPERDAY **3**
STOP LOSS **$1,500**	PROFIT TARGET **$800**
USEBREAKEVEN **false**	BREAKEVEN TRIGGER **$400**
NOISE FILTER **0.20**	

// VERIFIED IMPLEMENTATION NOTE

The code computes standard R2 in the [0, 1] range. With R2TakeProfitThreshold = 2.00, the R2 exit cannot trigger. Monetary target and stop remain active; break-even is disabled in the official inputs.

// FULL SAMPLE: 13/01/2017 - 31/08/2026

KEY RESULTS.

NET PROFIT	PROFIT FACTOR	WINNING TRADES	MAX DRAWDOWN
$265,780	1.23	69.54%	-$17,420

TRADES	AVERAGE TRADE	SHARPE RATIO	TIME IN MARKET
2,518	$105.55	0.44	28.99%

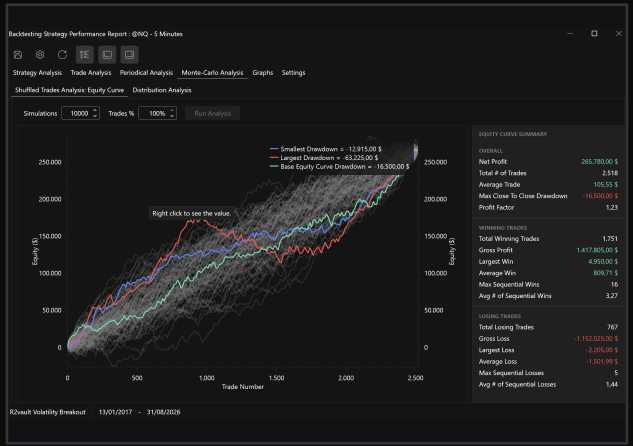
// ADJUSTED RESULT

The platform reports adjusted net profit of $190,300.45. This should be read alongside its commission and slippage assumptions.

// EQUITY AND RESAMPLED PATHS

EVIDENCE.





// OUT-OF-SAMPLE SEGMENT

15/01/2025 - 31/08/2026: 564 trades, $88,675 net profit, 1.37 profit factor and 71.45% profitable. No dedicated optimization was run on this segment.

// MONTE CARLO: 10,000 SIMULATIONS

The report shows -$12,915 smallest drawdown, -$63,225 largest drawdown, and -$16,500 base equity drawdown. It reshuffles historical trade order; it is not a future-risk forecast.

// DEVELOPMENT SAMPLE: 19/01/2017 - 31/12/2024

IN-SAMPLE TEST.

NET PROFIT **$180,635**	PROFIT FACTOR **1.20**
WINNING TRADES **69.08%**	MAX DRAWDOWN **-$17,420**
TRADES **1,947**	AVERAGE TRADE **$92.78**

// PURPOSE OF THE SAMPLE

Historical development interval. The 2025-2026 segment is treated separately, without dedicated optimization, to limit the risk of overfitting.

// FULL SAMPLE: 2017 - 2026

ANNUAL RETURNS.

2026	$55,500	2025	$29,645
2024	$32,525	2023	$28,015
2022	$13,085	2021	$37,000
2020	$32,580	2019	$11,980
2018	$5,145	2017	$20,305

// READING THE SERIES

Every calendar year in the full sample is positive. This is a historical observation, not evidence that future years will be profitable or that annual outcomes are independent.

// LIMITS AND NEXT STEPS

VERDICT.

R2Vault is a compact and testable proposition: weak linearity defines a local range, positive slope selects direction, and the proportional noise filter requires a more meaningful breakout.

The limited two-version development and absence of out-of-sample optimization are deliberate anti-overfitting choices. They reduce, but do not eliminate, model risk or regime risk.

// NEXT STEPS

Validate costs and fills, stress R2Length and noise sensitivity, formally preserve the in-sample/out-of-sample split, and observe forward simulation without ad hoc changes.

// DISCLAIMER

This document is descriptive, not financial advice. Historical and simulated results are not indicative of future results.